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Push past your resistance to strength training

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Using stretchy bands to build muscle can be an easy and effective way to burn more calories and boost your heart health.

Most people know that brisk walking and other movement that gets your heart beating faster is great for your cardiovascular health. But even health-conscious folks often overlook another key fitness recommendation: doing muscle-strengthening exercises at least twice a week.

Not crazy about the idea of going to a gym and hoisting heavy weights? There's no need to do so. You can work your muscles with stretchy bands (called resistance or exercise bands) at home. In fact, if you're out of shape or overweight, a simple resistance band workout can be a good way to ease into exercise, says Dr. Edward Phillips, assistant professor of physical medicine and rehabilitation at Harvard Medical School.

"Resistance training is inexpensive and fun, and it may help with weight loss. And for older, less-fit people, it can help build leg and core strength to make walking or other types of exercise possible," says Dr. Phillips. What's more, all types of strength training (including weight training done with free weights or machines) can benefit your heart and blood vessels in several different ways.

Chest press

Place the band around your back under your arms. Hold each end with your arms bent, elbows pointing out and hands by your armpits. Extend your arms straight in front of you at chest height. Slowly bend your arms and return to the starting position.

Exercise photos by Michael Carroll

Burning more calories

Boosting muscle mass speeds up metabolism, which helps people burn more calories, even at rest. "For every pound of untrained muscle you have, you spend about five or six calories per day just to maintain it. But if you train that muscle, its energy needs go up to about nine calories per pound per day," says Dr. Phillips. A faster metabolism also helps prevent weight gain.

Strength training causes tiny tears (called microtrauma) in the muscles. The body's effort to repair and remodel muscle cells boosts its energy requirements for 72 hours after a session of exercise. That's the reason behind the recommendation to do strength training on nonconsecutive days, Dr. Phillips explains. Muscle-building exercise also has been shown to lower body fat, improve cholesterol profiles, and help ward off diabetes—all of which help reduce the risk of heart disease.

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NEW RELEASE

Cardio Exercise: *7 workouts to boost energy, fight disease, and help you live longer*

www.health.harvard.edu/cardio

FIVE THINGS TO DO THIS MONTH

1 Substitute salmon for one meat meal per week. You'll avoid unhealthy saturated fat in favor of potentially heart-protecting fats. (page 3)

2 Family history of premature heart attacks? Ask your doctor about a test for lipoprotein(a). (page 4)

3 Identify your sex-specific heart risks. For women, they include pregnancy-related complications; for men, it's erectile dysfunction. (page 5)

4 Learn the causes of orthostatic hypotension. Warning signs include feeling dizzy and lightheaded after standing. (page 6)

5 Train your muscles with resistance bands. Use these stretchy elastic bands to strengthen your leg and core muscles (page 7).

Resistance training ... from p. 1

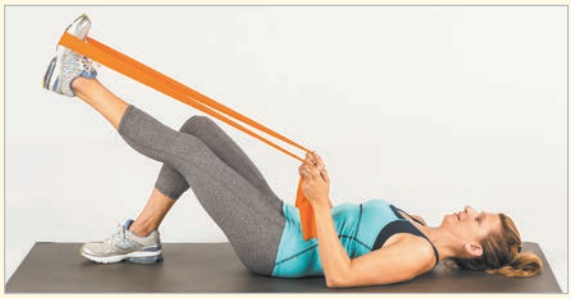
Band practice

Resistance bands or tubes come in a variety of resistance levels, designated by color. Bands look like big, wide rubber bands. Or you might prefer the version shaped like a tube—these have handles and tend to be more durable. Prices start at about $10 apiece, and many are sold in sets so you get a variety of resistance levels.

Start by trying the exercises featured on page 1 and below. To avoid injury and muscle soreness, take it easy at first. Use the band that gives the least resistance (the stretchiest one) and aim for a mid-range level of effort (say, about 5 or 6 on a scale of 1 to 10). Start with a single set of eight to 12 repetitions (reps) of each move, then gradually build up to two or three sets as you feel able. Rest for a minute or two between each set. Additional exercises and more detailed instructions are available in the Harvard Special Health Report *Strength and Power Training for All Ages* (www.health.harvard.edu/SPA).

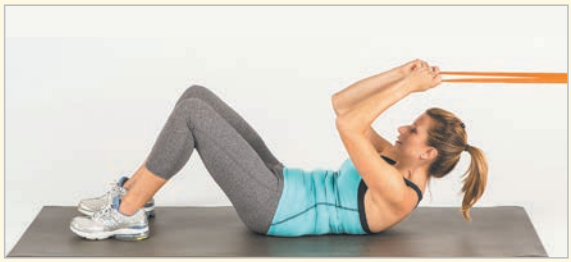
Leg press

Lie on your back on the floor with your knees bent and feet flat on the floor. Raise your left foot off the floor, bringing your knee toward your chest. Loop the band around the arch of your foot and hold an end of the band in each hand with your hands by your knee. Extend your leg, pressing your foot away from you. Slowly return to the starting position. This is one rep. Finish all reps, then repeat with your right leg. This completes one set.



Ab crunch with band

Loop the band around the leg of a heavy piece of furniture, a railing, or a pole about a foot above floor level. Grasp one end in each hand. Lie on your back with your head toward the anchor point, your knees bent, and your feet flat on the floor. Bend your arms and position your hands above your face. Tighten your abdominal muscles and lift your head, shoulders, and upper body.



Exercise photos by Michael Carroll

Vegetable of the month

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Leafy greens

A salad is a great way to meet your daily vegetable requirement—just toss a few of your favorite veggies together with lettuce or other greens. Most grocery stores sell a variety of salad greens (separate or mixed), prewashed and bagged for convenience. Arugula and watercress are peppery, dandelion and escarole are bitter, mâche and iceberg are sweet, and a green called tatsoi has a mustard flavor. Baby spinach is another popular option.

The USDA recommends two to three cups of vegetables per day for adults. But because greens aren't very dense, it actually takes about two cups of raw greens to make the nutritional equivalent of a one-cup serving of vegetables.

That's also true for sturdier leafy greens such as kale, chard, collards, and bok choy. But cooking reduces their volume by at least half. When tallying up your servings, count one cup of cooked greens as one serving.

Nutritional info: Leafy greens are rich in a combination of vitamins A, C, and K; several B vitamins; and potassium. They also contain fiber and are low in calories. As a general rule, the darker the green, the more nutrients it contains.

Easy recipe: Heat a tablespoon of olive oil in a large pan. Add one clove of minced garlic and a large bunch of coarsely chopped kale, spinach, chard, or other greens. Stir-fry until tender.

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